Proceedings of Machine Learning Research vol vvv:1–13, 2027

# Eddies: Dynamical Action Tokens with Certified Composition

**Aditya Narendra** ADITYA.NARENDRA@MBZUAI.AC.AE
**Grik Tadevosyan** GRIK.TADEVOSYAN@MBZUAI.AC.AE
**Abdalla Swikir** ABDALLA.SWIKIR@MBZUAI.AC.AE
*Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi, UAE*

## Abstract

Action tokens describe intended behaviour but do not quantify tracking accuracy or error propagation across skills. We introduce Eddies, action tokens that combine a primitive bound to a task frame, an identified robot-specific closed-loop model, and a conformal residual envelope. Our composer, Streamer, propagates tracking-error bounds across hand-overs, including the change in equilibrium; for fixed, stable models and exchangeable episodes this yields probabilistic tracking certificates for entire Eddie streams. Each Eddie gates its own commands to respect a tracking tolerance whenever its residual envelope holds. Online adaptation lets a new task start without its own calibration and controls the long-run miss rate. On ten simulated tasks across four robot configurations, Eddies produce tighter bounds under a common certification procedure than global models, per-primitive models and learned tokenizers. Certificates are violated in 5 of 2,000 simulated test episodes and in none of the real-robot cube-stacking runs (50 per certificate), and transfer across hands and physics engines shows that certificates depend on the executing system's dynamics. Code and videos: https://ozzey.github.io/papers/eddies/.

**Keywords:** Action tokens, certified composition, conformal prediction, system identification, manipulation

## 1. Introduction

Learned robot policies, vision-language-action models among them, act through tokens: binned or compressed codes that decode to reference commands (Brohan et al., 2023; Pertsch et al., 2025; Shafiullah et al., 2022; Lee et al., 2024), or primitives that a language model strings together (Ahn et al., 2022). This treats an action like a word, but an action is not a word: its effect depends on where the robot starts and which robot runs it, and the next action starts where the previous one left the robot. No action token bounds how well the robot tracks it or how its error carries to the next.

We make the token carry this information (Fig. 1). An *Eddie* (Embedding-space Discrete Dynamical Instruction with Envelope) is a primitive bound to a task frame, such as grasping a peg, plus a model of how this robot tracks the primitive and an envelope on the model's error. Our composer, Streamer, chains Eddies into an *Eddie stream*: every Eddie gates its own next command against the tracking tolerance, stopping the stream when the check fails, and the error bounds chain through hand-overs into a bound over the whole stream. The certificate covers tracking only; whether a skill succeeds is left to symbolic contracts and to the supervisor that picks the next Eddie.

Our contributions are as follows. (i) Eddies: action tokens that carry an identified closed-loop model and a conformal envelope, built for each robot from its own primitives (Section 4). On the Franka, splitting the dynamics per token roughly halves the certified tube of one global model or one model per primitive, and a skill's certificate can change with the robot and the physics engine that run it. (ii) A chaining rule that carries the actual bound across hand-overs, including the jump

© 2027 A. Narendra, G. Tadevosyan & A. Swikir.

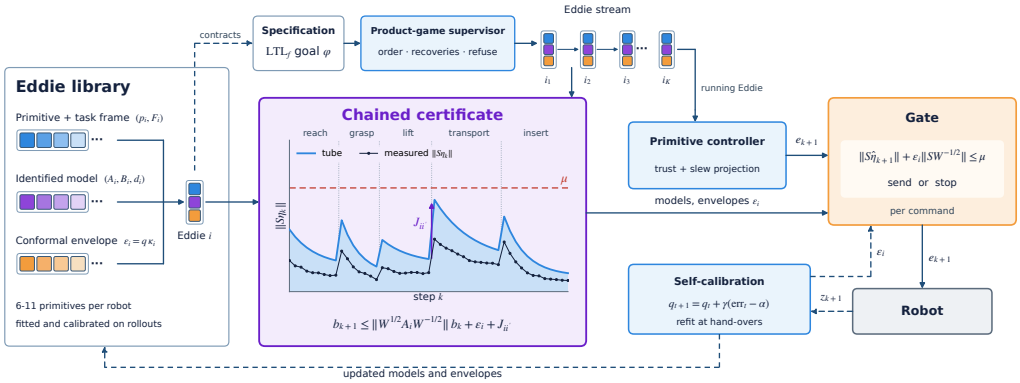



Figure 1: Streamer chains certified Eddies into an Eddie stream; each Eddie gates its own commands and is updated from the closed loop.

between equilibria, which certifies hand-overs even where funnel containment fails. (iii) A self-calibrating gate carried by each Eddie, which checks the Eddie's own next command and tunes itself from closed-loop feedback, with a one-step look-ahead guarantee when the envelope holds and a long-run miss-rate bound (Theorem 1). (iv) A comparison with learned action tokenizers under the same certificate, in which Eddies certify more tightly on all four robot configurations. We evaluate ten tasks on four robot configurations in MuJoCo, three of the tasks also in Isaac Sim, and cube stacking on a real Franka; proofs, identification and per-task tables are in the supplement.

## 2. Related Work

Action tokenizers have moved from per-dimension bins (Brohan et al., 2023) to richer codes based on k-means, frequency transforms, vector quantization and B-splines (Shafiullah et al., 2022; Pertsch et al., 2025; Lee et al., 2024; Mete et al., 2024; Zhou et al., 2025); none models how the robot tracks the action. Primitive libraries (Dalal et al., 2021; Nasiriany et al., 2022) and language-model planners over primitives (Ahn et al., 2022) name skills without a tracking model.

Conformal prediction gives coverage guarantees for prediction and reachability analysis (Lindemann et al., 2023; Cleaveland et al., 2024; Lindemann et al., 2025); combined with contraction analysis and Koopman models it yields probabilistic tracking-error bounds (Hirano and Tsukamoto, 2026; Wei et al., 2025), and Nath et al. (2026) reuse one Koopman model and one calibration across reference trajectories. Eddies attach models and envelopes to individual tokens and compose the bounds across hand-overs, and adaptive conformal inference (Gibbs and Candès, 2021) lets the envelopes adapt online when the task changes. Runtime monitors detect generative-policy failures from action consistency and task progress (Agia et al., 2024), but build no tracking-error tubes.

Funnel libraries compose controllers through containment conditions between successive funnels (Burridge et al., 1999; Tedrake et al., 2010; Majumdar and Tedrake, 2017), and temporal-logic imitation gives learned skills a related guarantee: mode-invariant, goal-reaching motion policies realize discrete plans (Wang et al., 2022). An Eddie stream is a switched affine system (Liberzon, 2003), whose stability is established by dwell-time conditions (Hespanha and Morse, 1999). These methods certify composition or stability; our chaining rule propagates a conformally calibrated tracking-error bound through the Eddie stream, including the jump between equilibria. At the task level, language-model planners (Ren et al., 2023; Wu et al., 2025) and strong fully observable non-deterministic (FOND) synthesis for goals in linear temporal logic on finite traces (LTL$_f$)

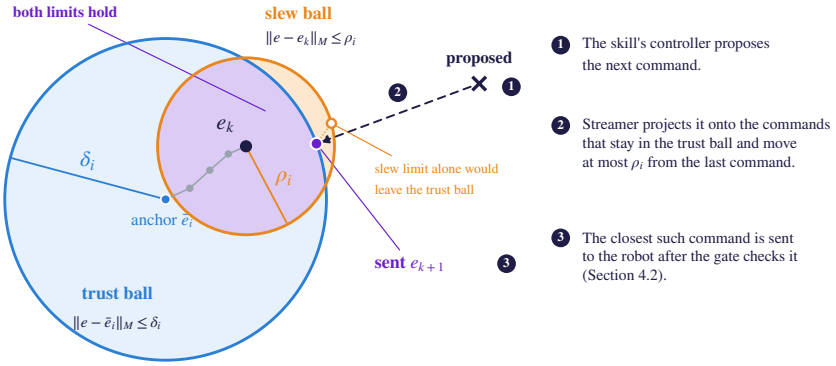



Figure 2: Each control step projects the proposed command onto the commands within the trust radius and the slew bound.

(Camacho and McIlraith, 2019) choose which skill executes next; Eddies complement that layer with tracking certificates for the chosen stream. Like a reference governor (Garone et al., 2017), the gate shapes only the command, but from each Eddie's chained tube; the conformal quantile adapts as in conformal decision theory (Lekeufack et al., 2024).

## 3. Problem Formulation

We consider manipulation in which a task is a sequence of skills whose commands a low-level controller tracks. We study the tracking error of the end-effector (the hand): the gap between the achieved hand pose and the closed loop's settling pose under the current command.

**Commands and measurements.** At step $k$, the controller receives a command $e_k \in \mathbb{R}^m$ that specifies the hand pose and the hand shape (gripper aperture or finger synergies), and a force or wrench limit if the controller accepts one. The measured output is the achieved hand pose $z_k \in \mathbb{R}^n$ (position and orientation), so hand shape and force limits enter only as inputs. Both $e_k$ and $z_k$ live in a normalized chart of the robot's task space, the embedding space that gives Eddies their name (Appendix B). Fixed diagonal weights $M \succ 0$ on $e$ and $W \succ 0$ on $z$, set once per robot, define the weighted norm $\|v\|_M = (v^\top M v)^{1/2}$. Without a subscript, $\|\cdot\|$ is the Euclidean or spectral norm.

**Skills and tasks.** Each robot has its own vocabulary of primitives (reach, grasp, insert, ...), shared by all its tasks (Table C.1). A skill is a primitive bound to a task frame. A task is a sequence of skills, which a supervisor chooses at run time from an $\mathrm{LTL}_f$ specification (Fig. 1). Switching skills is a hand-over. While skill $i$ runs, Streamer keeps every command within a trust radius $\delta_i$ of the skill's anchor $\bar{e}_i$, its nominal command, and limits the change per step to a slew bound $\rho_i$ (Fig. 2):

$$\|e_k - \bar{e}_i\|_M \le \delta_i, \qquad \|e_{k+1} - e_k\|_M \le \rho_i. \tag{1}$$

**Tracking error.** Let $r_k \in \mathbb{R}^n$ be this settling pose, as predicted by the running skill's model (Section 4). The tracking error at step $k$ is $\eta_k = z_k - r_k$. $S \in \mathbb{R}^{3\times n}$ selects the position rows of $z$, scaled to metres, so $\|S\eta_k\|$ is the hand-position error, which the task tolerates up to $\mu$.

**Problem.** Given a robot, its primitive vocabulary and a tolerance $\mu$, construct, for every sequence of skills whose commands satisfy (1), an upper bound on the hand-position error $\|S\eta_k\|$ that (i) holds at a prescribed coverage level at every step of the sequence, including across hand-overs; (ii) can be evaluated before execution and checked at every step during execution, so that the sequence is

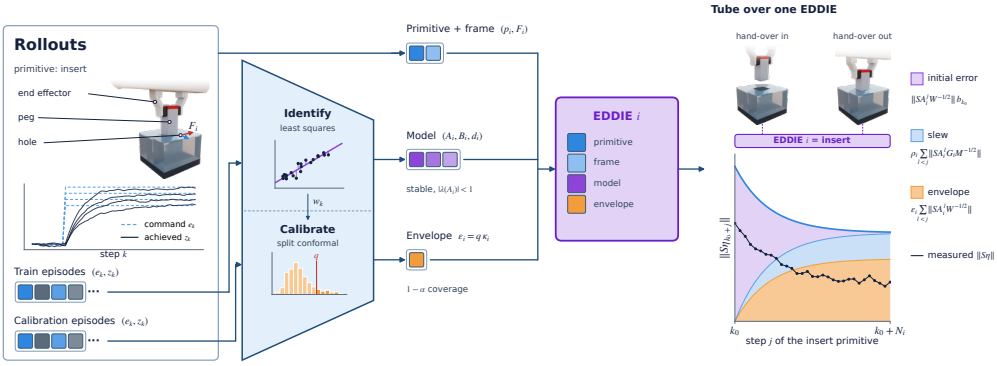



Figure 3: An Eddie pairs a model identified from rollouts with an envelope from split conformal calibration; together they give its tube.

stopped before the error exceeds $\mu$; and (iii) can be learned for a new task without a calibration set for that task. Whether a skill achieves its goal is outside the scope of this problem.

## 4. Eddies

### 4.1. What an Eddie carries

**Definition 1** *An Eddie $i$ is a primitive bound to a frame (e.g. a grasp bound to the peg), together with an affine closed-loop model $(A_i, B_i, d_i)$ of how the robot tracks it and an envelope $\varepsilon_i \geq 0$ on the model's residual, all in one token: while $i$ runs,*

$$z_{k+1} = A_i z_k + B_i e_k + d_i + w_k, \qquad \|W^{1/2} w_k\| \leq \varepsilon_i. \tag{2}$$

The model describes the closed loop, robot plus low-level controller: $A_i \in \mathbb{R}^{n\times n}$ is the state-transition matrix (its eigenvalues set how fast the hand settles), $B_i \in \mathbb{R}^{n\times m}$ how the command moves the next pose, $d_i \in \mathbb{R}^n$ a constant offset such as gravity or a steady contact force, and $w_k \in \mathbb{R}^n$ the residual, the part of $z_{k+1}$ that the model does not explain. The envelope is not a hard bound: it holds with probability at least $1-\alpha$ over episodes (3).

Fig. 3 shows how an Eddie is built: its model is fitted by least squares on the robot's training episodes that use it, regularized toward a well-excited reference Eddie of the same robot (Appendix B). If $A_i$ is stable, a constant command $e$ settles at $z_i^\star(e) = (I - A_i)^{-1}(B_i e + d_i)$. With $i(k)$ the Eddie running at step $k$, the settling pose of Section 3 is $r_k = z^\star_{i(k)}(e_k)$, so the tracking error is $\eta_k = z_k - z^\star_{i(k)}(e_k)$. On the Franka, this equilibrium lies a median 0.2 cm from the command itself (at most 1.92 cm), so a bound on $\|S\eta_k\|$ bounds the error to the command up to this offset, and the error to the command stays within the Eddie's tube on 99.9% of its test steps (Appendix D).

The envelope comes from split conformal prediction. Each Eddie carries a normalizer $\kappa_i > 0$ (the 90th percentile of its largest residual over training episodes); an episode's score is its largest normalized residual, $s = \max_k \|W^{1/2} w_k\| / \kappa_{i(k)}$, so one quantile $q$ calibrated on whole episodes covers every Eddie of the stream. For a miscoverage level $\alpha \in (0,1)$ and $N$ calibration episodes with sorted scores $s_{(1)} \leq \cdots \leq s_{(N)}$, the envelopes are

$$\varepsilon_i = q\,\kappa_i, \qquad q = s_{(\lceil (N+1)(1-\alpha) \rceil)}. \tag{3}$$

If a task's episodes are exchangeable, a new episode keeps every residual within its envelope with probability at least $1-\alpha$ (Vovk, 2012; Cleaveland et al., 2024) (Fig. 4a).

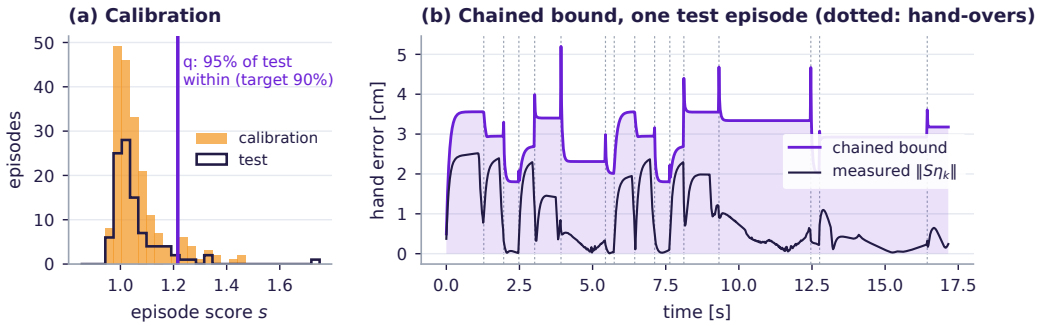



Figure 4: (a) Staged insertion: $q$ of (3) keeps 95% of the first 100 test episodes in the envelope (94% of 200; target 90%). (b) One episode: the chained bound of Theorem 1 (violet) stays above the measured error, jumping by $J_{ii'}$ at hand-overs (dotted).

On this event, unrolling the error dynamics of the model (Hirano and Tsukamoto, 2026; Wei et al., 2025) bounds the tracking error $j$ steps after Eddie $i$ starts at step $k_0$:

$$\|S\eta_{k_0+j}\| \le \|SA_i^j W^{-1/2}\|\,\|W^{1/2}\eta_{k_0}\| + \rho_i \sum_{l<j} \|SA_i^l G_i M^{-1/2}\| + \varepsilon_i \sum_{l<j} \|SA_i^l W^{-1/2}\|, \qquad (4)$$

with $G_i = (I - A_i)^{-1} B_i$ the steady-state gain. The terms are the decay of the initial error, the lag behind a command moving up to $\rho_i$ per step, and the accumulated residual. The coverage statement after (3) and the bound (4) are standard; Appendix E proves them (Lemma E.1, Proposition E.2).

### 4.2. The Eddie's gate and chaining

At a hand-over the error does not simply carry over: it also jumps by the gap between the two Eddies' equilibria. Each Eddie therefore checks every command before it is sent, jump included, with its current model and the envelope $\varepsilon_i = \max(q_t, s)\,\kappa_i$, where $q_t$ is the conformal quantile of episode $t$, shared by the task's Eddies, $s \ge 0$ is the episode's score so far, which widens the envelope once it exceeds $q_t$, and $\kappa_i$ is refit with the model. Before sending $e_{k+1}$, the running Eddie $i$ predicts the next tracking error with its own model and the equilibrium of the Eddie $i'$ that will run step $k+1$. Inside an Eddie $i' = i$, so the next Eddie enters only at a hand-over:

$$\begin{aligned} &\hat{\eta}_{k+1} = A_i z_k + B_i e_k + d_i - z^\star_{i'}(e_{k+1}), \\ &\text{send } e_{k+1} \text{ only if} \quad \|S\hat{\eta}_{k+1}\| + \varepsilon_i \,\|SW^{-1/2}\| \le \mu. \end{aligned} \qquad (5)$$

The gate also stops the robot if the measured error $\|S\eta_k\|$ exceeds $\mu$; a stopped robot holds its last command, and the run ends unfinished unless the check passes again (Section 5.5).

Every Eddie starts from the most specific prior available (its own calibrated certificate, else its primitive's, else one model of the whole robot), and at each hand-over Algorithm 1 refits its model toward this prior by ridge regression on the task's data so far, kept only if stable (Appendix B). After every episode, the algorithm moves the conformal quantile by the quantile form of adaptive conformal inference (Gibbs and Candès, 2021; Angelopoulos et al., 2023):

$$q_{t+1} = q_t + \gamma\,(\mathrm{err}_t - \alpha), \qquad \mathrm{err}_t = \mathbf{1}\{s_t > q_t\}, \qquad (6)$$

where the step size is $\gamma = 0.5$ and $s_t$ is episode $t$'s score under the current models and $\kappa_i$.

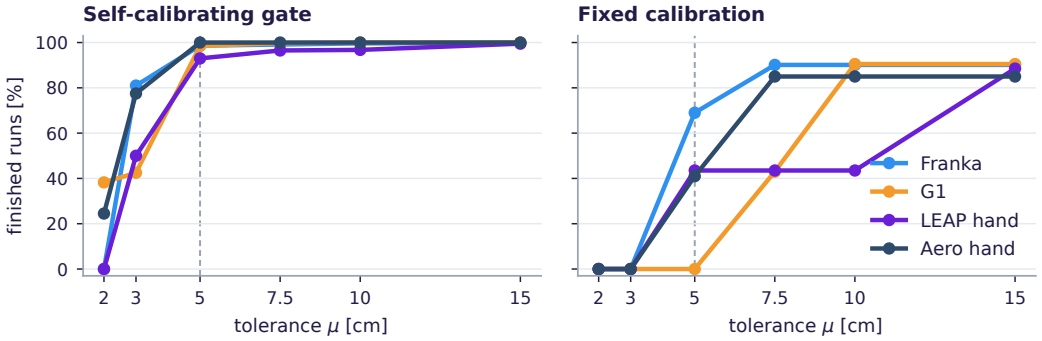



Figure 5: Finished runs against tolerance for both gates from the task's own calibration; one curve per robot (200 test episodes per task).

**Algorithm 1:** Self-calibrating gate, episode $t$

**Input:** models and normalizers of the Eddies (priors), conformal quantile $q_t$, tolerance $\mu$
$s \leftarrow 0$;
**for** *each step $k$, with running Eddie $i$* **do**
  **if** $\|S\eta_k\| > \mu$ ***or*** (5) *fails* **then** stop the robot;
  send $e_{k+1}$ and measure $z_{k+1}$;
  $s \leftarrow \max\left(s,\ \|W^{1/2}w_k\|/\kappa_i\right)$; `// episode score so far`
  $\varepsilon_i \leftarrow \max(q_t, s)\,\kappa_i$; `// widen the envelope if needed`
  **if** *a hand-over follows* **then** refit $(A_i, B_i, d_i, \kappa_i)$ by ridge regression toward its prior;
**end**
$q_{t+1} \leftarrow q_t + \gamma\,(\mathbf{1}\{s > q_t\} - \alpha)$; `// adaptive conformal update`

Our guarantees rest on two assumptions. **(A1)** *Exchangeable episodes:* the episodes of a task come from independent scene draws, while steps within an episode may be correlated. **(A2)** *Stable models:* every model the gate uses has $\max_\lambda |\lambda(A_i)| < 1$. We certify only Eddies that satisfy it.

**Theorem 1 (Certified Eddie streams)** *Let (A2) and (1) hold. Let $b_k = \|W^{1/2}\eta_k\|$ be the weighted tracking error and $a_i = \|W^{1/2}A_iW^{-1/2}\|$ the one-step gain of Eddie $i$. Let $J_{ii'} = \|W^{1/2}\big(z_i^\star(e_k) - z_{i'}^\star(e_{k+1})\big)\|$ be the jump between the equilibria when Eddie $i$ hands over to Eddie $i'$ after step $k$.*

*(a)* Chained certificate. *Under (A1) and with the envelopes (3), with probability at least $1-\alpha$ over episodes, at every step: within each Eddie the error stays within the tube (4) from its value when the Eddie starts, and across each hand-over*

$$b_{k+1} \le a_i\, b_k + \varepsilon_i + J_{ii'}. \tag{7}$$

*Chained from the measured error at the first step, these bounds hold for $\|S\eta_k\|$ over the whole Eddie stream and are finite for every finite stream.*

*(b)* Gate. *For any stable models, without (A1): whenever the residual of the running Eddie stays within its envelope, the command its gate (5) sends keeps $\|S\eta_{k+1}\| \le \mu$, and every run that finishes stays within $\mu$.*

*(c)* Miss rate. *Under (6), with scores and $q_1$ in $[0, s_{\max}]$, $|\frac{1}{T}\sum_{t\le T}\mathrm{err}_t - \alpha| \le (s_{\max}+\gamma)/(\gamma T)$ for every $T$ and any sequence of episodes and tasks.*

Part (a) needs no contraction across hand-overs (81 of 85 certified Eddies have $a_i < 1$; Appendix B) and assumes the fixed models of (3); (b) and (c) hold for any stable models, refitted ones included (Proposition E.3); (b) is the gate's design guarantee, checked before each command.

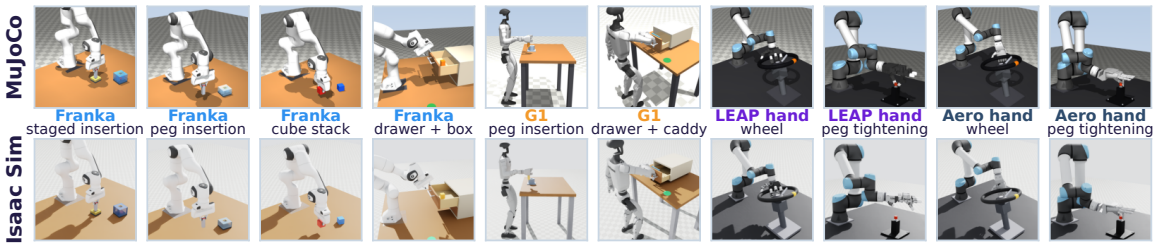



Figure 6: The ten tasks in MuJoCo (top) and Isaac Sim (bottom; three certified there) from the same task code; the LEAP and Aero hands sit on a UR5e arm.

### 4.3. Vocabulary and specifications

The Eddies of a task share one calibrated conformal quantile; a primitive added later gets a fixed share of the error budget, so adding it never changes an existing certificate (Proposition E.4; Section 5.4). Each primitive also carries a symbolic contract (precondition and possible outcomes). Given an $\mathrm{LTL}_f$ goal (De Giacomo and Vardi, 2013), Streamer solves a product game over the Eddies; its strategy is the Eddie stream, with recoveries; the goal is refused before motion if none exists, and every Eddie on every branch goes through the gate (Appendix A).

## 5. Experiments

We test Eddies on ten MuJoCo tasks with four robot configurations (Todorov et al., 2012; Zakka et al., 2022). Each task uses only its robot's primitives (Fig. 6). The Franka does peg insertion, cube stacking, drawer opening with a box transfer, and staged insertion (16 Eddies: pick the peg, stage it on a fixture, regrasp, insert at 1–2 mm clearance, push to full depth). The G1 humanoid does peg insertion and drawer opening with a caddy transfer, and a UR5e with a LEAP hand and with an Aero hand each turns a wheel and tightens a peg. Staged insertion, peg insertion and cube stacking are also certified in Isaac Sim from the same code (NVIDIA, 2026).

Each task has 200 training, 200 calibration and 200 test episodes, $\alpha = 0.1$, and a certificate is violated if the measured hand position leaves its tube. Gates are replayed on the logged test episodes, so the outcome of every stopped run without the gate is known.

### 5.1. Certificates on four robot configurations

The certificates hold: with fixed calibration, tubes are violated in 5 of 2,000 test episodes, and most are 3–5 cm wide (Table D.1). Envelopes are missed more often, in 6–16.5% of Franka test episodes (Table D.3; the drawer's 16.5% is within sampling noise, Appendix D), and in 8.5–10.5% under the self-calibrating gate, close to $\alpha$. At 5 cm, the recursion of Theorem 1 certifies 77–100% of the hand-overs of every Franka task, against 69–86% for funnel containment; whole episodes certify a priori on peg insertion at 5 cm and on three tasks from 7.5 cm (Table D.4).

Checking one command at a time more than doubles the finished runs (Fig. 5): at 5 cm with fixed calibration, checking every planned Eddie's tube before motion finishes 890 of 2,000 runs, checking only the next command 1,968, and the self-calibrating gate, which adds the refits and the updates of $q$, 1,955 (98%) from the same calibration; its benefit shows on new tasks (Section 5.4). At 2 cm, 1,627 runs would exceed the tolerance $\mu$, and the self-calibrating gate stops 97% of them before they do, where a threshold on the measured error alone stops none in time; at 3 cm it stops 387 of 395 and 1,328 runs finish (Appendix D). The fixed gate checks each planned tube before motion

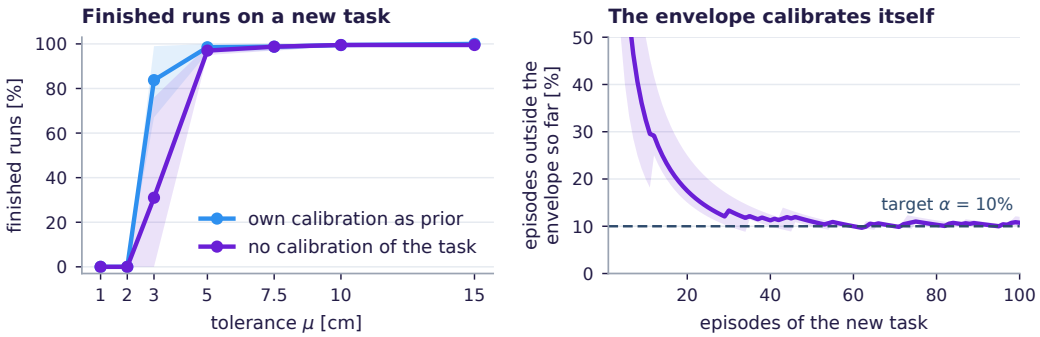



Figure 7: Four Franka tasks held out of training, no calibration (left: also own as prior): finished runs (left), share of episodes so far outside the envelope (right); mean and range.

Table 1: Our certificate on different tokens of each robot's data: BeT, FAST, VQ-BeT and a global model as in Nath et al. (2026), BeT and FAST with as many tokens as the robot has Eddies (35 Franka, 13 per dexterous hand, 25 G1). Without a known token order, the check before motion covers every token a method emits on the task. Tube p90 [cm] and runs finished of 400 test runs (Franka: first 100 of each task), with fixed calibration, at $\mu = 5$ / 7.5 cm (G1: 7.5 / 10 cm; its balance control widens the tubes to a p90 of up to 7.9 cm, so no plan fits 5 cm). Bold: not significantly below the best (paired bootstrap, 95% intervals of about $\pm 10$ runs; Appendix D).

| | Franka | | LEAP hand | | Aero hand | | G1 | |
|---|---|---|---|---|---|---|---|---|
| Tokens | tube | finished 5 / 7.5 cm | tube | finished 5 / 7.5 cm | tube | finished 5 / 7.5 cm | tube | finished 7.5 / 10 cm |
| **Eddies** | **4.3** | **275 / 359** | **11.9** | **174 / 174** | **6.5** | **164 / 340** | **7.9** | **172 / 362** |
| One global model | 8.5 | 0 / 262 | 22.0 | 0 / 0 | 24.4 | 0 / 0 | 79.3 | 0 / 0 |
| One model per primitive | 8.1 | 0 / 264 | 12.5 | 0 / 0 | 12.0 | 0 / 170 | 15.7 | 0 / 0 |
| BeT | 8.7 | 169 / 261 | 26.4 | 0 / 0 | 12.6 | 0 / 0 | 9.3 | 0 / 187 |
| FAST + k-means | 13.7 | 177 / 272 | 55.3 | 0 / 0 | 16.2 | 0 / 0 | 14.9 | 0 / 0 |
| VQ-BeT | 16.0 | 264 / 264 | 18.2 | 0 / 0 | 21.0 | 0 / 0 | 10.2 | 0 / 0 |
| Eddies, one reference model | 18.4 | 0 / 0 | 17.6 | 0 / 0 | 10.5 | 0 / 0 | 60.1 | **172** / 172 |
| Eddies, per-step bound | – | 0 / 0 | – | 0 / 0 | 8.7 | 0 / 0 | – | 0 / 0 |

and each hand-over by (7); the one-step check lets runs finish where these do not fit (Table D.4). With every Eddie 1.5–4 times faster, the tasks still succeed, but 35–100% of runs would leave $\mu$ at 5 cm; with fixed calibration, the gate refuses every Eddie stream before motion (Table D.5).

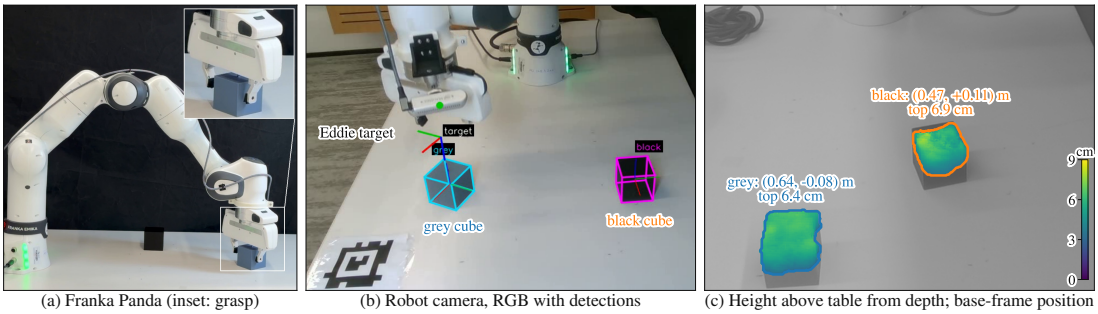
### 5.2. Eddies against other tokens

Published tokenizers carry no model or envelope, so we apply our certification procedure to each on the same data (Table 1); any difference in tightness comes from the token. Eddies are tightest on every robot and finish the most runs; on the dexterous hands and the G1 no other tokens finish a run at the smaller tolerance. Token counts are matched; with about twice the tokens, BeT comes level with Eddies on the Franka at twice the hand-overs (Appendix D).

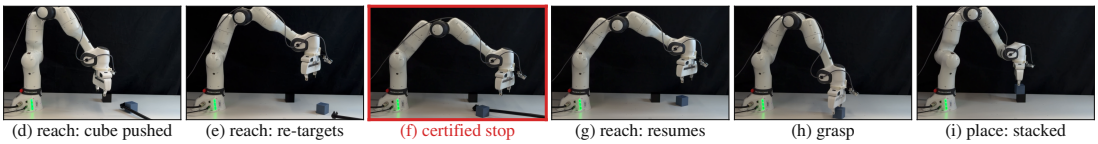
Calibration and each Eddie's own model both matter: a per-step envelope with a union bound (Hirano and Tsukamoto, 2026) certifies no Eddie stream at these tolerances (too few calibration steps), and one reference model for all Eddies widens the tube 1.5–7.6 times (Appendix D).

Table 2: Certificates borrowed from a source system or re-identified on the target (Isaac Sim: refit on the MuJoCo data plus 10 Isaac Sim episodes per task, calibrated on 50 more); viol.: test episodes that leave a tube; finished: runs the gate finishes at $\mu = 15$ cm; bold: the better per task, both when equal.

| | | | borrowed | | re-identified | |
|---|---|---|---|---|---|---|
| Source → target | Task | episodes | viol. | finished | viol. | finished |
| LEAP → Aero | Wheel | 200 | 20 | 78 | **0** | **176** |
| | Peg tightening | 200 | 5 | 0 | **0** | **164** |
| Aero → LEAP | Wheel | 200 | **0** | 22 | 1 | **180** |
| | Peg tightening | 200 | **0** | 0 | **0** | **174** |
| MuJoCo → Isaac Sim | Staged insertion | 50 | 10 | 22 | **0** | **45** |
| | Peg insertion | 50 | 5 | 19 | **0** | **46** |




(a) Franka Panda (inset: grasp) (b) Robot camera, RGB with detections (c) Height above table from depth; base-frame position



(d) reach: cube pushed (e) reach: re-targets (f) certified stop (g) reach: resumes (h) grasp (i) place: stacked

Figure 8: Cube stacking on the real Franka. (a) The arm grasping the grey cube (inset). (b) Robot-camera RGB feed with the detected cubes and the active Eddie's target. (c) Cube heights from the camera's depth image. (d–i) A stick pushes the cube and the arm re-targets; out of the safe set, the arm holds in a certified stop until the cube is back, then grasps and stacks it.

### 5.3. Certificates across robots and engines

Table 2 compares certificates borrowed from another system (LEAP/Aero; MuJoCo/Isaac Sim, Fig. C.1) with re-identified ones: borrowed ones are violated or too wide to finish most runs, re-identified ones are violated at most once and finish over 80%, in Isaac Sim after ten episodes.

### 5.4. New tasks, primitives and specifications

A new task reuses the Eddies it shares with other tasks, falls back to primitive or global models for the rest and tunes itself as it runs (Fig. 7; Table D.2). With no task episodes, the self-calibrating gate finishes nearly every run at 5 cm with 1–5 false stops; the conformal certificate finishes none without calibration and needs about 55 episodes to finish 83–93. At 2 cm a new task's gate stops 17 and 37 of 100 cube-stacking and peg-insertion runs in time (91 and 99 with its own calibration).

With a new push primitive, every staged-insertion Eddie certifies, and the push leaves the nine existing primitives' certificates unchanged. From $\mathrm{LTL}_f$ goals, the product game recovers all four manually specified Franka Eddie streams, runs five new goals (99–100% success) and refuses three impossible ones before motion; a network trained on the supervisor's choices (92–100% of held-out hand-overs right) emits the Eddies with the supervisor's 99–100% nominal success (Appendix D).

Table 3: Cube stacking in MuJoCo (first 100 test episodes), Isaac Sim and on the real Franka at $\mu = 5$ cm. Viol.: runs that leave a tube; tube p90: 90th percentile of the tube radius; fin./stop.: runs without / with a stop; caught: runs at 2–4× speed refused or stopped before leaving $\mu$, of those that would.

| | | | | | | fixed gate | | self-calibrating gate | |
|---|---|---|---|---|---|---|---|---|---|
| System | Certificate | runs | success | viol. | tube p90 [cm] | fin. / stop. | caught | fin. / stop. | caught |
| MuJoCo | MuJoCo | 100 | 99 | 0 | 4.3 | 89 / 11 | 298 / 298 | 96 / 4 | 287 / 298 |
| Isaac Sim | MuJoCo | 50 | 50 | 0 | 4.3 | 31 / 19 | 150 / 150 | 48 / 2 | 148 / 150 |
| | re-identified | 50 | 50 | 0 | 4.0 | 41 / 9 | 150 / 150 | 48 / 2 | 147 / 150 |
| robot | MuJoCo | 50 | 50 | 0 | 4.3 | 30 / 20 | 150 / 150 | 46 / 4 | 145 / 150 |
| | re-identified | 50 | 50 | 0 | 4.1 | 43 / 7 | 150 / 150 | 45 / 5 | 143 / 150 |

### 5.5. Real experiments

Cube stacking (eight Eddies) also runs in Isaac Sim and on a real Franka Panda (Fig. 8), where the 1 kHz compliant controller of CRISP (San José Pro et al., 2026) tracks the 50 Hz Eddie commands. Both gates see the same runs (same seeds), replayed on the logs in simulation and online on the robot, where only the fixed gate acts and half of the 50 runs per certificate are perturbed; the certificate is MuJoCo's or refit with 50 robot episodes and calibrated on 50 more (Table 3).

In contrast to the insertion tasks, the MuJoCo certificate holds on both systems: no certificate is violated, and every run stacks the cube; its envelopes are missed more often, so the fixed gate stops 19 of 50 runs in Isaac Sim and 20 on the robot (9 and 7 once re-identified), the self-calibrating gate 2–5; stopped runs resume once the check passes and still stack. At 2–4× speed (ungated, gates replayed), the fixed gate catches every run that would leave $\mu$ and the self-calibrating gate 95–99%.

A cube pushed during the approach is followed by the arm; pushed out of the safe set (cube poses for which the reach Eddie's check (5) passes), it triggers a certified stop that holds the arm until the cube is back (15 s here), then stacks it (Fig. 8d–i).

## 6. Conclusion

Eddies are action tokens that carry their robot's dynamics, chain into a whole-task certificate, gate their own commands and self-calibrate. They certify more tightly than other tokens, hold on a real Franka, and transfer shows that certificates follow the executing system's dynamics. Limitations: the certificate bounds position tracking, not orientation or contact outcomes, and $\mu$ is not a task outcome; per-episode guarantees need fixed calibration (the self-calibrating gate, only logged on the robot, bounds the long-run miss rate); models are affine per Eddie. Next steps are nonlinear or contact-aware models, orientation in the certificate and the self-calibrating gate acting on the robot. A learned proposer already emits Eddies (Section 5.4); a vision-language-action model could too, under the same guarantee.

**Generative AI use.** Generative AI tools were used for writing and editing. The authors reviewed and verified all AI-assisted content and take responsibility for the final content of the paper.

## References

Christopher Agia, Rohan Sinha, Jingyun Yang, Ziang Cao, Rika Antonova, Marco Pavone, and Jeannette Bohg. Unpacking failure modes of generative policies: Runtime monitoring of con-

sistency and progress. In *Conference on Robot Learning (CoRL)*, volume 270 of *Proceedings of Machine Learning Research*, pages 689–723, 2024.

Michael Ahn, Anthony Brohan, Noah Brown, Yevgen Chebotar, et al. Do as I can, not as I say: Grounding language in robotic affordances. In *Conference on Robot Learning (CoRL)*, volume 205 of *Proceedings of Machine Learning Research*, pages 287–318, 2022.

Anastasios N. Angelopoulos, Emmanuel J. Candès, and Ryan J. Tibshirani. Conformal PID control for time series prediction. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.

Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. In *Conference on Robot Learning (CoRL)*, volume 229 of *Proceedings of Machine Learning Research*, pages 2165–2183, 2023.

Robert R. Burridge, Alfred A. Rizzi, and Daniel E. Koditschek. Sequential composition of dynamically dexterous robot behaviors. *The International Journal of Robotics Research*, 18(6):534–555, 1999.

Alberto Camacho and Sheila A. McIlraith. Strong fully observable non-deterministic planning with LTL and LTL$_f$ goals. In *International Joint Conference on Artificial Intelligence (IJCAI)*, pages 5523–5531, 2019.

Matthew Cleaveland, Insup Lee, George J. Pappas, and Lars Lindemann. Conformal prediction regions for time series using linear complementarity programming. In *AAAI Conference on Artificial Intelligence*, volume 38, pages 20984–20992, 2024. doi: 10.1609/aaai.v38i19.30089.

Murtaza Dalal, Deepak Pathak, and Ruslan Salakhutdinov. Accelerating robotic reinforcement learning via parameterized action primitives. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, 2021.

Giuseppe De Giacomo and Moshe Y. Vardi. Linear temporal logic and linear dynamic logic on finite traces. In *International Joint Conference on Artificial Intelligence (IJCAI)*, pages 854–859, 2013.

Emanuele Garone, Stefano Di Cairano, and Ilya Kolmanovsky. Reference and command governors for systems with constraints: A survey on theory and applications. *Automatica*, 75:306–328, 2017. doi: 10.1016/j.automatica.2016.08.013.

Isaac Gibbs and Emmanuel Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 1660–1672, 2021.

João P. Hespanha and A. Stephen Morse. Stability of switched systems with average dwell-time. In *IEEE Conference on Decision and Control (CDC)*, pages 2655–2660, 1999.

Koki Hirano and Hiroyasu Tsukamoto. Conformal Koopman for embedded nonlinear control with statistical robustness: Theory and real-world validation. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2026. arXiv:2603.21580.

Seungjae Lee, Yibin Wang, Haritheja Etukuru, H. Jin Kim, Nur Muhammad Mahi Shafiullah, and Lerrel Pinto. Behavior generation with latent actions. In *International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pages 26991–27008, 2024.

Jordan Lekeufack, Anastasios N. Angelopoulos, Andrea Bajcsy, Michael I. Jordan, and Jitendra Malik. Conformal decision theory: Safe autonomous decisions from imperfect predictions. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 11668–11675, 2024. doi: 10.1109/ICRA57147.2024.10610041.

Daniel Liberzon. *Switching in Systems and Control*. Birkhäuser, Boston, 2003.

Lars Lindemann, Matthew Cleaveland, Gihyun Shim, and George J. Pappas. Safe planning in dynamic environments using conformal prediction. *IEEE Robotics and Automation Letters*, 8(8): 5116–5123, 2023. doi: 10.1109/LRA.2023.3292071.

Lars Lindemann, Yiqi Zhao, Xinyi Yu, George J. Pappas, and Jyotirmoy V. Deshmukh. Formal verification and control with conformal prediction: Practical safety guarantees for autonomous systems. *IEEE Control Systems Magazine*, 45(6):72–122, 2025. doi: 10.1109/MCS.2025.3611545.

Anirudha Majumdar and Russ Tedrake. Funnel libraries for real-time robust feedback motion planning. *The International Journal of Robotics Research*, 36(8):947–982, 2017. arXiv:1601.04037.

Atharva Mete, Haotian Xue, Albert Wilcox, Yongxin Chen, and Animesh Garg. QueST: Self-supervised skill abstractions for learning continuous control. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, 2024.

Soroush Nasiriany, Huihan Liu, and Yuke Zhu. Augmenting reinforcement learning with behavior primitives for diverse manipulation tasks. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 7477–7484, 2022. doi: 10.1109/ICRA46639.2022.9812140.

Devesh Nath, Haoran Yin, and Glen Chou. Scalable data-driven reachability analysis and control via Koopman operators with conformal coverage guarantees. In *Learning for Dynamics and Control Conference (L4DC)*, volume 331 of *Proceedings of Machine Learning Research*, pages 71–97, 2026.

NVIDIA. Isaac Sim 6.0 release notes. https://docs.isaacsim.omniverse.nvidia.com/6.0.0/overview/release_notes.html, 2026. Accessed Sep. 19, 2026.

Karl Pertsch, Kyle Stachowicz, Brian Ichter, Danny Driess, Suraj Nair, Quan Vuong, Oier Mees, Chelsea Finn, and Sergey Levine. FAST: Efficient action tokenization for vision-language-action models. In *Robotics: Science and Systems (RSS)*, 2025. doi: 10.15607/RSS.2025.XXI.012.

Allen Z. Ren, Anushri Dixit, Alexandra Bodrova, Sumeet Singh, Stephen Tu, Noah Brown, et al. Robots that ask for help: Uncertainty alignment for large language model planners. In *Conference on Robot Learning (CoRL)*, volume 229 of *Proceedings of Machine Learning Research*, pages 661–682, 2023.

Daniel San José Pro, Oliver Hausdörfer, Ralf Römer, Maximilian Dösch, Martin Schuck, and Angela P. Schoellig. CRISP – compliant ROS2 controllers for learning-based manipulation policies and teleoperation. *IEEE Robotics and Automation Practice*, 1:103–107, 2026. doi: 10.1109/RAP.2026.3690413.

Nur Muhammad Mahi Shafiullah, Zichen Jeff Cui, Ariuntuya Altanzaya, and Lerrel Pinto. Behavior transformers: Cloning $k$ modes with one stone. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, 2022.

Russ Tedrake, Ian R. Manchester, Mark Tobenkin, and John W. Roberts. LQR-trees: Feedback motion planning via sums-of-squares verification. *The International Journal of Robotics Research*, 29(8):1038–1052, 2010.

Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 5026–5033, 2012. doi: 10.1109/IROS.2012.6386109.

Vladimir Vovk. Conditional validity of inductive conformal predictors. In *Asian Conference on Machine Learning (ACML)*, volume 25 of *Proceedings of Machine Learning Research*, pages 475–490, 2012.

Yanwei Wang, Nadia Figueroa, Shen Li, Ankit Shah, and Julie Shah. Temporal logic imitation: Learning plan-satisficing motion policies from demonstrations. In *Conference on Robot Learning (CoRL)*, volume 205 of *Proceedings of Machine Learning Research*, pages 94–105, 2022.

Sihang Wei, Melkior Ornik, and Hiroyasu Tsukamoto. Conformal contraction for robust nonlinear control with distribution-free uncertainty quantification. In *IEEE Conference on Decision and Control (CDC)*, pages 3502–3507, 2025. doi: 10.1109/CDC57313.2025.11312813.

Yi Wu, Zikang Xiong, Yiran Hu, Shreyash S. Iyengar, Nan Jiang, Aniket Bera, Lin Tan, and Suresh Jagannathan. SELP: Generating safe and efficient task plans for robot agents with large language models. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

Kevin Zakka, Yuval Tassa, and MuJoCo Menagerie contributors. MuJoCo Menagerie: A collection of high-quality simulation models for MuJoCo. https://github.com/google-deepmind/mujoco_menagerie, 2022.

Hongyi Zhou, Weiran Liao, Xi Huang, Yucheng Tang, Fabian Otto, Xiaogang Jia, et al. BEAST: Efficient tokenization of B-splines encoded action sequences for imitation learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 38, 2025. arXiv:2506.06072.